



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 03.08.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : **EEE G559**
Course Title : **ADV Power Electronics**
Instructor-in-Charge : **Rajneesh Kumar**
Instructor(s) :
Tutorial/Practical Instructors : **Shubham Priyadarshi**

1. Course Description: This course covers advance power electronic circuits which includes DC-DC converters, Rectifiers and Inverters. Steady state analysis of these converters is discussed in detail with emphasis on proper selection of devices and magnetic components. Mathematical modelling of converter transfer function using various approaches is also included for controller design of these converters. Experiments are designed to understand the working of these power electronic circuits supported with MATLAB/SIMULINK based simulation platform.

2. Scope and Objective of the Course:

- Understand the main elements of power electronic circuits*
- Steady-state analysis of basic power electronic circuits like DC-DC converters, rectifiers and inverters*
- Derivation of AC small signal model and transfer functions of power converters*
- Experimental validation Basic power converter circuits*
- Development of one new converter for selected applications*

3. Textbooks:

T1: Robert W. Erickson, Dragan Maksimovic, *Fundamental Power Electronics*: 2nd edit. Springer International, 2001

T2: N. Mohan, T. Undeland, and W. Robbins, *Power Electronics: Converters, Applications, and Design*, 3rd edit. John Wiley & Sons, 2003

4. Reference Books: Class notes and Journal articles

5. Course Plan:

Module No.	Lecture Session	References	Learning outcomes
1. Elements of Power Electronics	L0-L4	T1(1.1-1.3, 4.1-4.3)	Efficient uses of power semiconductor switches and realization in converter circuits. Switching losses and converter efficiency analysis
2. DC-DC converters: Buck, Boost, Buck Boost and derived converters	L5-L12	T1(2.1-2.3, 3.1-3.3, 3.5, 5.1-5.4, 6.3.1, 6.3.4)	DC-DC converter steady analysis and voltage conversion ratio in CCM and DCM. Converter component design and selection for a given voltage and current ripple
3. Converter Dynamics and transfer function	L13-25	T1(7.1-7.6, 8.2, 11.1-11.4)	AC equivalent circuit modeling in CCM and DCM. Transfer function of some basic CCM converters and significance of pole zero location
4. Modern rectifiers and Power System Harmonics	L26-35	T1(16.1-16.6, 17.1-2, 18.1-2, 18.4-6, 18.7)	Understanding working principles and applications of Line commutated rectifiers and pulse-width modulated rectifiers
5. Basic of Inverters	L36-40	T2(8.1-7)	Working principles of single phase and three phase Inverters using various PWM schemes



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90 Min.	20	Refer Timetable	Open Book
Comprehensive Examination	3 h	30	Refer Timetable	Close Book/Open Book
Assignments	-----	30	-----	-----
Laboratory experiments	-----	20	-----	-----

7. Chamber Consultation Hour: Will be announced.

8. Notices: on Nalanda

9. Make-up Policy: As per AUGSD guidelines

10. Note (if any):

Rajneesh Kumar
Instructor-in-charge
Course No. EEE G559